

ER Diagram

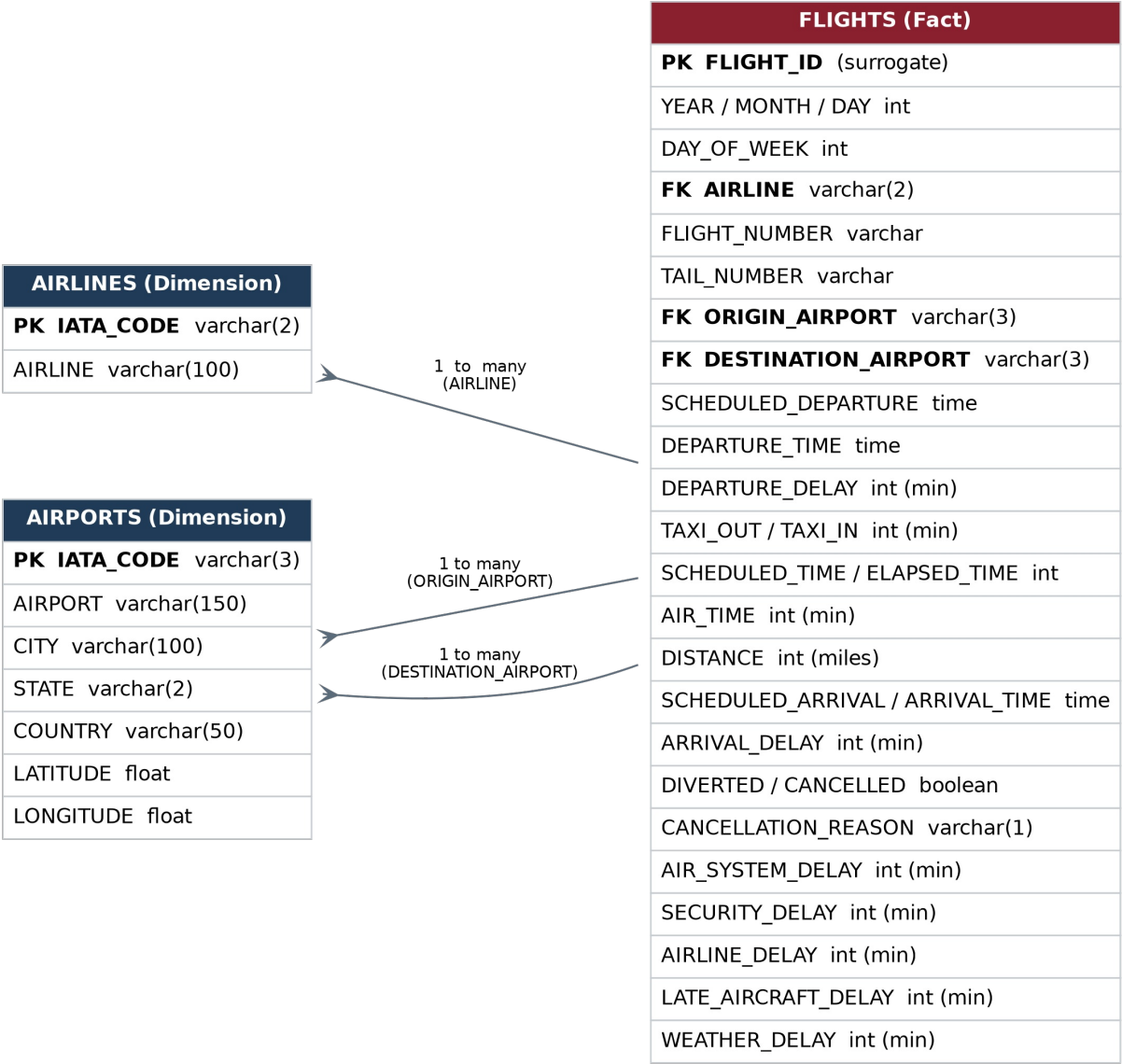
Table Relationships — Airline Flight Delay Analysis

Data Source	Maven Analytics — Airline Flight Delays (U.S. DOT, 2015)
Schema Type	Star schema (1 fact table, 2 dimension tables)
Version	1.0
Date	August 2026

1. Schema Overview

The Airline Flight Delays dataset forms a simple **star schema**: the `flights` table is the central fact table, and `airlines` and `airports` are dimension (lookup) tables. Notably, `airports` is referenced **twice** from the fact table — once for the origin airport and once for the destination airport — a common pattern called a "role-playing dimension."

2. Entity Relationship Diagram



3. Relationship Definitions

Relationship	Cardinality	Join Key	Description
<code>airlines</code> → <code>flights</code>	One-to-Many	<code>airlines.IATA_CODE</code> = <code>flights.AIRLINE</code>	Each airline operates many flights; each flight is operated by exactly one airline.
<code>airports</code> → <code>flights</code> (as Origin)	One-to-Many	<code>airports.IATA_CODE</code> = <code>flights.ORIGIN_AIRPORT</code>	Each airport can be the departure point of many flights.
<code>airports</code> → <code>flights</code> (as Destination)	One-to-Many	<code>airports.IATA_CODE</code> = <code>flights.DESTINATION_AIRPORT</code>	Each airport can be the arrival point of many flights.

4. Modeling Notes for BI Tools

Role-playing dimension handling: Because `airports` joins to `flights` twice (origin and destination), most BI tools (Power BI, Tableau) require either two separate relationships with one marked "inactive" and activated via DAX (`USERELATIONSHIP`), or — the simpler and recommended approach for this project — importing the airports table **twice** as two distinct tables: `Origin Airports` and `Destination Airports` , each with a single active one-to-many relationship to the fact table.

5. Primary & Foreign Keys Summary

Table	Primary Key	Foreign Keys
<code>airlines</code>	<code>IATA_CODE</code>	None (dimension)
<code>airports</code>	<code>IATA_CODE</code>	None (dimension)
<code>flights</code>	None declared (surrogate row ID recommended: <code>FLIGHT_ID</code>)	<code>AIRLINE</code> → <code>airlines.IATA_CODE</code> <code>ORIGIN_AIRPORT</code> → <code>airports.IATA_CODE</code> <code>DESTINATION_AIRPORT</code> → <code>airports.IATA_CODE</code>

The raw flights CSV does not include a natural single-column unique identifier; a surrogate key (row number or composite of YEAR-MONTH-DAY-AIRLINE-FLIGHT_NUMBER) should be generated during data load if a unique row identifier is required by the BI tool or downstream joins.

6. Cardinality Legend

Symbol	Meaning
Crow's foot (many) → single tick (one)	One dimension record relates to many fact records — the standard star-schema relationship used throughout this model.